Document Control

Title

Midazolam Injection in Adults
Standard Operating Procedure

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Consulted with the following stakeholders: • DTC	**Contact responsible for implementation and monitoring compliance:** Director of Pharmacy/ Clinical Pharmacy Manager **Education/ training will be provided by:** Workforce Development

Approval and Review Process

- Drugs and Therapeutics Committee

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CONTENTS

Document Control 1
1. Background 3
2. Purpose 3
3. Scope 4
4. Location 4
5. Equipment 4
6. Procedure 4
Supply and Storage of Midazolam Injection 4
Prescribing and administration of Midazolam Injection 5
Monitoring 5
Incidents/Adverse Drug Reactions 5
7. References 6
8. Associated Documentation 6

1. Background

1.1. Midazolam is a benzodiazepine, which is used for conscious sedation and palliative care purposes.

1.2. The National Patient Safety Agency (NPSA) issued a Rapid Response Report (NPSA/2008RRR011) in December 2008 to advise that a significant number of adverse events and three deaths have occurred due to parenteral administration of an overdose of midazolam in adults.

1.3. The procedure specifically deals with the use of midazolam injection in adults. Where guidance on sedation in general is required please refer to the appropriate Sedation Guidelines.

1.4. This procedure does not affect the supply of other midazolam formulations.

1.5. For Buccal Midazolam please refer to Buccal Midazolam Standard Operating Procedure (Buccal Midazolam for the control of epileptic seizures within the Salaried Dental Service) | Policies

2. Purpose

2.1. The Standard Operating Procedure (SOP) has been written to ensure that parenteral midazolam is ordered, stored, prescribed and administered safely for conscious sedation, general anaesthesia and palliative care. This will be achieved by:

- Restricting the strengths of midazolam ampoules that can be stored in different clinical areas
- Ensuring that the reversal agent flumazenil is available in all clinical areas that stock midazolam ampoules

3. Scope

3.1. This Standard Operating Procedure (SOP) relates to the following staff groups who may be involved in the assessment and delivery of in-patient care on the wards:

- Registered practitioners

4. Location

- This Standard Operating Procedure for Midazolam Injection in Adults can be implemented in all clinical areas where competent staff are available to undertake this role.
- Staff undertaking this procedure must be able to demonstrate continued competence as per the organisations policy on assessing and maintaining competence.

5. Equipment

- Appropriate supplies of medication

6. Procedure

Supply and Storage of Midazolam Injection

Midazolam is a Schedule 3 Controlled Drug. The ordering, storage, prescribing and administration must follow Northern Devon Healthcare Trust Policy and Standard Operating Procedures:

- Medicines Policy and associated Standard Operating Procedures
- Controlled Drugs Standard Operating Procedure
- Administering Injectable Medicines Standard Operating Procedure
- Preparing Injectable Medicines Standard Operating Procedure

Midazolam can only be supplied by supplying Pharmacies on a valid controlled drug order.

The following strengths are available to order from supplying pharmacies:

- **1 mg/ml** ampoules N.B. check volume available from the supplying pharmacy
- **2 mg/ml in 5ml** ampoules
- **5 mg/ml in 2 ml** ampoules

Storage restrictions for midazolam ampoules held as stock:

- **1 mg/ml** ampoules must be stocked in clinical areas where it is used for conscious sedation.
- **2 mg/ml** ampoules can only be stocked in clinical areas for the purpose of general anaesthesia and palliative care.
- **5 mg/ml** ampoules can only be stocked in clinical areas for the purpose of general anaesthesia and palliative care.

Flumazenil Injection 100 micrograms per ml, for reversal of midazolam effects, must be stored in all areas where midazolam is used.

Prescribing and administration of Midazolam Injection

Prescribers should ensure they are familiar with midazolam dosage regimes and that where necessary are familiar with relevant Sedation Guidelines.

When prescribing or administering Midazolam for palliative care purposes the Joint Formulary should be followed and the Northern Devon Healthcare Trust End of Life Care Pathway and Just In Case Bag /Box Standard Operating Procedure should be followed. Syringe Drivers should be used in accordance with the Northern Devon Healthcare Trust McKinley Syringe Driver policy.

The registered professional administering injectable midazolam is responsible for ensuring that the appropriate strength of midazolam is used to prepare injections.

The manager in charge of each clinical area/ward is responsible for ensuring that flumazenil injection is available to reverse the effects of midazolam when required.

It is the responsibility of the manager in charge of each clinical area/ward to ensure that staff know where to find flumazenil in an emergency situation.

Monitoring

The following information will be reviewed quarterly to ensure compliance with this procedure and to ensure that flumazenil use is not the result of accidental over sedation with midazolam.

- Flumazenil issues from supplying pharmacy
- Midazolam (all strengths) issued from supplying pharmacy

Incidents/Adverse Drug Reactions

In the event of a medication incident or an adverse drug reaction involving midazolam immediate care including the administration of flumezanil according to the British National Formulary(BNF) will need to be given to minimise harm to the patient.

The patient’s medical / surgical team / GP should be informed in addition to the prescriber if this is someone different.

The incident must be recorded in the patient record indicating the actions taken.

The incident must be reported in accordance with Northern Devon Healthcare Trust Incident reporting policy.

Report a serious adverse drug reaction to the Medicines and Healthcare products Regulatory Agency (MHRA) through the yellow care reporting system – details are contained in the British National Formulary (BNF) and at www.mhra.gov.uk . This can be undertaken by staff or the patient if this is appropriate.

7. References

- NPSA Rapid Response Report - NPSA/2008/RRR011 Reducing risk of overdose with midazolam injection in adults – Issued December 2008
  http://www.nrls.npsa.nhs.uk/resources/?entryid45=59896&q=0%c2%acmidazolam%c2%ac
- Medicines and Healthcare procedures Regulatory Agency (MHRA)
  http://www.mhra.gov.uk/index.htm#page=DynamicListMedicines
- British National Formulary (BNF)
  BNF.org

8. Associated Documentation

8.1. Northern Devon Healthcare NHS Trust Policies for :

Northern Devon Healthcare NHS Trust Policies for:

- Medicines Policy
  Medicines Policy « Northern Devon Healthcare NHS Trust NDHT
- Controlled Drugs Standard Operating Procedures
- Incident and Management Investigation Policy
- Medication Errors – Standard Operating Procedure (Managing and Supporting Staff Following a Medication Error) | Policies
- Consent Policy
  http://ndht.ndevon.swest.nhs.uk/policies/?p=171
- Injectable Medicines Policy (Prescribing, Preparing and Administering Injectable Medicines Policy)
  http://ndht.ndevon.swest.nhs.uk/policies/?p=9138
- Waste Management Procedure
- Salaried Dental Services policy and procedures for Intravenous Sedation
- Syringe Driver Policy
  McKinley Syringe Driver Pumps | Policies
- End of Life Policy